Topological Ring Detection for Market Manipulation

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Abstract—Wash trading and laundering show up as directed loops among a few accounts, not as unusual single transactions. We ask whether topological features of those loops add predictive value over graph-baseline features (B) and direction and behaviour features (D, which include the ring's birth value). Candidate rings are enumerated exactly inside strongly connected components, and persistent homology of a directed flag complex of each ring's neighbourhood supplies the topological features (T). The comparison B+D+T versus B+D was fixed before the experiments and is judged by paired grouped-bootstrap intervals. On the IBM AML benchmark (itself synthetic), D carries the signal and T adds no detectable lift: for logistic regression on HI-Small the paired PR-AUC difference is +0.104 [−0.002, +0.171] with 14 positive groups, and every interval on LI-Small (10 positive groups) also includes 0. Two of six transfer cells between the datasets are negative. On a synthetic generator nearly separable by D, a small lift appears in 7 of 12 condition-model cells, but not where the design predicted. The ring-birth score that works on HI-Small (0.314) does not replicate on LI-Small (0.006). In simulation, bandit thresholds lose to periodic re-tuning, while online gradient ascent reaches 0.981 of the hindsight oracle. A real-data NFT experiment was not run.

Index Terms—wash trading, anti-money laundering, persistent homology, directed flag complex, graph anomaly detection, negative results

I. Introduction

A wash trade moves nothing of economic value. Colluding accounts pass an asset or money around until the last recipient hands it back to the first, and the recorded volume goes up. Measurement studies find this on decentralized exchanges and in NFT markets [1]–[3], and a recent working paper reports it on a prediction-market exchange [4]. Laundering through layering has the same shape, with value cycling through intermediaries. Every payment in such a scheme looks ordinary. What gives it away is the loop: a small group of accounts in which value circulates and nets to about zero for each member.

Two gaps motivate the work. Detectors built around dense subgraphs, communities or node centrality look for tightly connected groups, yet a ring can be sparse, with each member trading with only two or three others. Cycle enumeration finds such loops but returns a yes or no, with no measure of strength, stability or type. Separately, a flagging threshold tuned once on history loses accuracy when manipulators change ring size, timing and decoy trades.

Persistent homology offers a graded description of a loop. In a directed flag complex a circular triangle $a \to b \to c \to a$ is never filled, so its loop persists, while a transitive triangle $a \to b$, $b \to c$, $a \to c$ is filled at once [5]. That makes the construction a natural fit for rings, and we set the experiment up to give it a fair chance. Our question is deliberately narrow: given the ring's direction and behaviour features, do topological features add predictive value? Ring birth, the largest edge weight on the ring, is a function of the ring's own edges, so we count it in the direction and behaviour set D, not in the topological set T. The primary comparison, fixed in advance, is B+D+T against B+D, where B are graph-baseline features. The IBM data (itself synthetic) hold only 10 to 14 positive groups, so we report the intervals as they are and tune nothing to narrow them.

Scope. We target the closed cycle: wash-trading rings and laundering cycles of up to six accounts. Pump-and-dump coordination is a temporal motif, not a cycle, and appears only as a small synthetic track without topology. Spoofing is out of scope because it is an order-book phenomenon with no signal in settled transactions. We evaluate on the IBM AMLworld benchmark, which is synthetic, and on our own synthetic generator with planted rings; no real market data were used.

Contributions.

1) A suspicion-weighted filtration and a directed-flag-complex pipeline for candidate rings, with exact account-level localization by cycle enumeration, two persistence backends that agree within tolerance, and an exact test of whether a ring's own loop is ever filled. We claim no priority for applying directed flag complexes to transaction graphs.
2) A pre-specified comparison of B+D+T and B+D

under group-aware cross-validation. On IBM AML data we find no detectable topological lift within either dataset, and two of six transfer cells are negative; D clearly beats B.

3) On a synthetic generator nearly separable by D, a small topological lift in 7 of 12 condition-model cells, not concentrated where the design predicted and absent on IBM; and a ring-birth signal that does not replicate across the IBM datasets (0.314 on HI-Small, 0.006 on LI-Small).
4) Exploratory and negative results, documented as such: bandit thresholds that lose to periodic retuning, a severity regression that recovers generator structure, weak clustering by pattern type, and small sensitivity to the infinite-bar cap (at most 0.024 in PR-AUC among the pre-set rules).

Section II places the work among existing detectors, Section III describes the pipeline, Section IV the data and protocol, Section V the results, and Section VI the limitations, including what was not done.

II. Related Work

Wash-trade measurement and rules. Empirical studies identify wash trading by rules over accounts and trades. On two order-book decentralized exchanges, Victor and Weintraud find a lower bound of 159 million USD wash-traded, more than 30% of tokens affected, and structures that mostly involve one or two accounts [1]. On seven NFT collections a repurchase rule (a wallet buys back a token within 30 days of selling it) flags on average 0.14% of transactions [3], while linking NFT ownership traces to the full Ethereum transaction network finds wash trading up to 24% of volume, with one outlier collection at 93% [2]. For stock markets, Cao et al. model collusion as a directed trader graph over matched orders and solve two knapsack-like steps by dynamic programming [6]. Sirolly et al. assume wash traders form nearly closed clusters of counterparties and apply an iterative network procedure to a prediction market; their paper is a working paper and not peer reviewed [4]. These methods see self-trades, short cycles and closed clusters. None assigns a ring a continuous strength from a filtration.

Graph and learning detectors for fraud and AML. Dense-subgraph detection under camouflage is addressed by Fraudar on bipartite review graphs [7]. For laundering, Weber et al. release a labelled Bitcoin graph and report that random forest outperforms graph convolutional networks [8]; Ouyang et al. classify laundering groups with subgraph contrastive learning [9]; Pei et al. embed subgraphs by structural roles and score them with Isolation Forest, restricting themselves to undirected, unweighted networks [10]. Chen et al. shrink strongly connected components and run a parallel depth-first search to mark suspicious accounts [11]. Learned detectors need labels and tuning and give little per-loop interpretation; component-and-search methods localize loops but return a binary answer. We use classical classifiers only.

Cycle enumeration. We localize rings by enumerating simple directed cycles up to a length bound with NetworkX [12], whose documentation names Johnson's algorithm for the unbounded case [13] and a version of the algorithm of Gupta and Suzumura for the bounded case [14]. A recent preprint presents examples in which the latter fails to enumerate valid cycles [15]; we did not test our enumeration against it, so our candidate recall may be affected in a way unrelated to the parameters we vary.

Topology on graphs. Ofori-Boateng et al. track the shape of multilayer blockchain graphs with clique persistent homology and report gains over state-of-the-art baselines for network-level anomalies [16]. A clique complex ignores direction, and it describes the network as a whole, not individual rings. Directed flag complexes and their persistent homology are computed by flagser, developed for brain-network data [5]; tournaplexes refine them by filling non-transitive cliques and distinguish graph dynamics more finely on the same kind of data [17]. Persistence diagrams must be vectorized for classifiers; we use persistence landscapes [18] with fixed settings, and note that learned weighting of persistence summaries exists [19] but was not tried. Our contribution here is a controlled test of what such features add for per-ring classification, not a new topological construction.

Drift and thresholds. Streaming fraud detection often uses drift detectors to decide when to retrain [20]. Bandits for changing reward distributions include discounted UCB [21] and sliding-window Thompson sampling [22], whose analysis has since been revisited and corrected in a preprint [23]. We simulate only, on one synthetic drift stream.

Data. The IBM AMLworld generator provides laundering data with complete labels [24]; SynthAML is a different synthetic AML benchmark built from real bank data [25], which we did not use. Temporal-motif analyses of cryptocurrency graphs show that whole-network counts can mislead and that per-user and per-period views reveal more [26], which is one reason we work with time windows and local neighbourhoods.

Where this work sits. Across these families a loop is either found and reported as binary, or described by network-level or node-level statistics. We keep exact enumeration as the localizer, because the persistence library interface we use cannot name the accounts of a loop, and ask a narrower question about the features: does a per-ring persistence description help once the ring's own weights and behaviour and the surrounding graph are already known? Three design choices keep the answer interpretable: ring birth sits in D, the baseline sees at least as much of the graph as T does, and near-copies of a ring are grouped so that they never straddle a split. We ran none of the other methods named above and make no

claim about them.

III. Method

The pipeline has three jobs: build a weighted graph from raw transactions, find candidate rings, and describe each ring in a form a classifier can use. Figure 1 summarizes it.

A. Windows, scores and weights

Time is cut into windows (5 days with a 1-day stride on IBM data). Within a window all payments from a to b form one directed edge carrying total volume V_{ab}, count, and the coefficient of variation (CV) of inter-payment gaps. Three scores lie in $[0, 1]$, higher meaning more suspicious. Pair reciprocity is $s_{\text{rec}} = 1 - |V_{ab} - V_{ba}|/(V_{ab} + V_{ba})$. Node flow balance is $s_{\text{bal}} = \min(\beta(a), \beta(b))$ with $\beta(x) = 1 - |\text{in}_x - \text{out}_x|/(\text{in}_x + \text{out}_x)$. Timing regularity is $s_{\text{reg}} = 1/(1+\text{CV})$; pairs with fewer than three payments have no CV and receive the window median. Each score is replaced by its rank within the window, scaled to $[0, 1]$, with ties taking the lowest rank. The edge weight is $w = 1 -$ mean of the three ranked scores, so a lower weight is more suspicious, matching the sublevel-set filtration in which low-weight cells enter first.

Proposition 1. In a ring of three or more accounts with no reciprocal pair, the ranked s_{rec} is 0 on every edge, so under equal weights $w \geq 1/3$ however balanced and regular the ring is. A multi-party ring cannot look maximally suspicious under equal weights, which motivates learning the combination weights on the probability simplex (Appendix A).

B. Candidate rings

A candidate generator keeps edges with $w \leq \tau$ (a quantile of the window's weights: 0.05 on IBM, 0.25 on synthetic data), computes strongly connected components, and enumerates simple directed cycles of length at most 6, with a cap of 20,000 cycles per component whose hits are recorded. Each cycle is an explicit set of accounts and edges. Its birth is the largest edge weight on it. De-duplication, as implemented, removes nothing for maximum length up to 9, so near-duplicate rings all remain candidates; the evaluation therefore groups them (Section IV). Localization is combinatorial because the pyflagser interface we use returns persistence pairs but no representative cycles.

C. Directed flag persistence

For a candidate we take the ring plus its one-hop neighbours in the sparsified graph and build the weighted directed flag complex up to dimension 2: vertices at 0, directed edges at their weight, and a 2-cell for each ordered triple (a, b, c) with edges $a \to b$, $a \to c$, $b \to c$ at the largest of the three weights. H0 and H1 diagrams come from flagser (float32) or a pure-Python $\mathbb{Z}/2$ reduction; the two must agree within tolerance, and tests check the canonical cases (Fig. 2). The diagram of the whole sparsified window is computed once per window.

What persistence can add beyond enumeration. For an isolated ring in a sparse neighbourhood the H1 class is born at the ring's birth value and never dies, so persistence repeats what enumeration and birth already give. It adds information only when the neighbourhood interacts with the loop. In Fig. 2(c) account d fills the circular triangle through three transitive triangles and the loop dies at the fill weight; chords can split a long ring into shorter cycles; and many incidental loops can crowd it. The synthetic generator plants these cases (dense, chorded and shortcut-filled scenarios) so that the method is tested where topology should matter, and the results are stratified by scenario for that reason.

D. Feature sets

T (311 columns) contains T_{local} (156 columns) from the ring neighbourhood and T_{win} (155 columns, identical for all rings of a window): three persistence landscapes [18] of 50 grid points on $[0, 1.1]$, counts of H1 and H0 classes, the number of infinite H1 bars (counted before capping), total persistence and persistence entropy. Infinite bars receive death equal to the window's maximum weight. T_{local} also holds an exact $\mathbb{Z}/2$ test, T_local_ring_inf, of whether the ring's own loop lies in the span of the boundaries of the directed 2-cells; if not, it survives. D (17 columns) contains ring length, birth, edge weights, the mean and minimum of each ranked score along the ring, member degree imbalance, and volume and timing statistics. B (12 columns) contains member degrees, betweenness (sampled with 100 sources on IBM windows, exact up to 3000 nodes), clustering, neighbourhood weight variance, density and size, all on the full window graph, so B sees at least as much of the graph as T. Feature builders never receive labels, planted structures or attempt identifiers, and a test checks this.

E. Labels

A transaction backs an edge of a structure if it is one of that structure's own transactions. A candidate is positive if at least 80% of its edges are backed by one structure (on IBM, one laundering attempt of any pattern type), background if none are, and ambiguous otherwise. On IBM a candidate whose flagged edges rest only on laundering rows that belong to no attempt is also ambiguous. The primary variant excludes ambiguous candidates; a second variant counts them positive.

F. Classifiers and threshold policies

Logistic regression (L2, $C = 1$), random forest (300 trees) and an RBF SVM ($C = 1$, gamma scale) with balanced class weights and fixed hyperparameters, each preceded by a scaler fitted in the training fold. No hyperparameter search of any kind was run. For the threshold study, a candidate is flagged when its score is

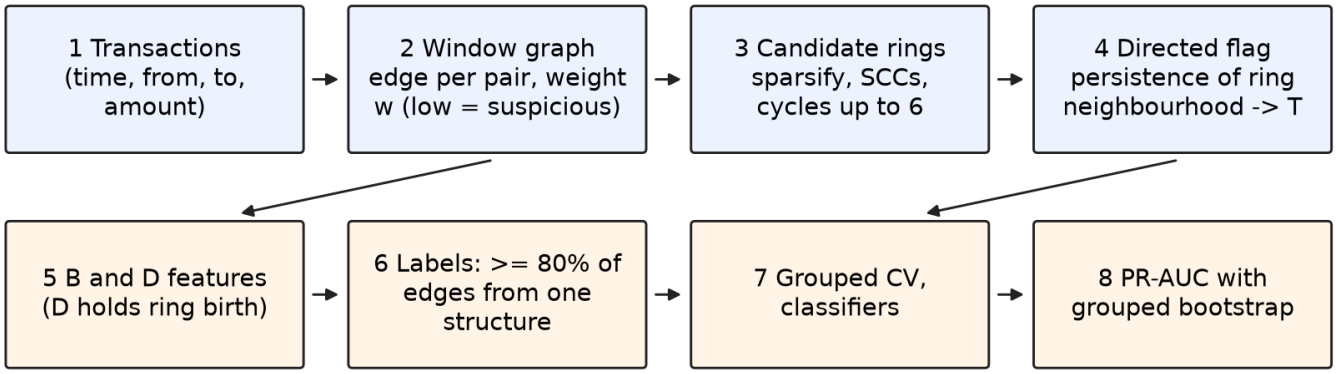


Fig. 1. Pipeline from transactions to out-of-fold PR-AUC. Blue: graph and topology stages; orange: features, labels and evaluation. The compared feature sets are B, B+D, B+D+T and T; the pre-specified primary comparison is B+D+T against B+D.

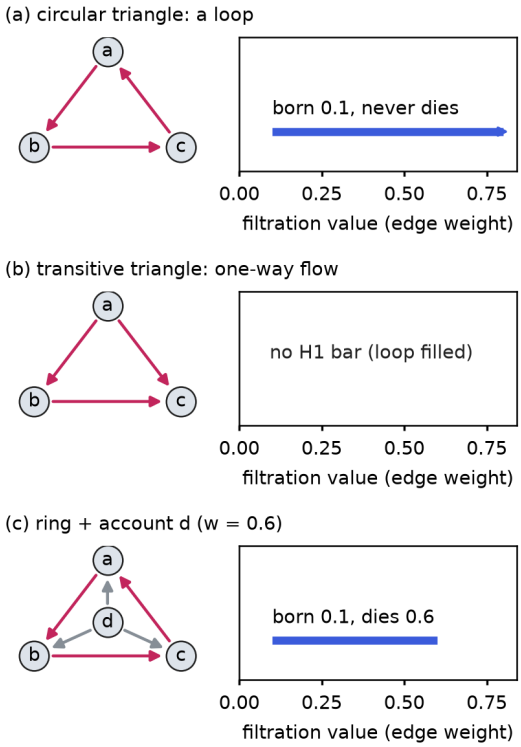


Fig. 2. H1 persistence bars computed by the project's reference backend on the three toy graphs. (a) A circular triangle with edge weights 0.1: one bar born at 0.1 that never dies (drawn to the cap). (b) A transitive triangle: filled, no H1 bar. (c) The same circular triangle plus account d paying all three members at weight 0.6: the loop dies at 0.6. Schematic; no data.

at most τ, with reward $\mathrm{TP} - \lambda\,\mathrm{FP}$ per window and labels arriving d windows late. Policies are: a threshold fixed after the first windows; re-tuning every M windows on labelled history with the same lag; online gradient ascent on a sigmoid-smoothed reward; discounted UCB [21]; sliding-window Thompson sampling [22]; and a hindsight oracle as an upper bound.

IV. Experimental Setup

The experiments are designed so that a null result is informative: the key comparison is fixed in advance, and near-duplicate rings never straddle a train-test split.

A. Data

IBM AMLworld. We use the HI-Small and LI-Small datasets of the AMLworld generator (Section II), which are synthetic but have complete labels, and its patterns file that groups laundering transactions into attempts of a given type. After dropping self-loops (11.6% of HI-Small rows) there are 4,487,133 HI and 6,119,572 LI transactions. Amounts are used in their own currency, so volume ratios mix currencies. Of the 5,177 laundering rows in HI-Small, 1,968 belong to no attempt. Laundering patterns need not be reciprocal, regular or balanced, so IBM validates the pipeline, not the wash-trading weights; Section V reports an audit of this.

Synthetic generator. A heavy-tailed background plus planted structures: isolated, dense-neighbourhood and chorded wash rings (positives); transitive decoys and shortcut-filled pseudo-rings, whose loop is filled by a fourth account (topology-sensitive negatives); and arbitrage loops, market makers and exchange hot wallets (hard negatives). Six seeds are pooled for each of four conditions (hard negatives off or on, crossed with the default dense scenario or one without chords among a ring's own accounts). The designer of the generator decides how separable the classes are, so synthetic results describe this generator only. With the default setting, 7 of 24 planted dense rings keep an open loop; with internal chords off, 18 of 24 do. A planted ring whose loop is filled keeps its positive label.

Table I gives candidate counts. The HI primary variant has 7555 candidates after excluding ambiguous ones.

B. Protocol

Rings recur across overlapping windows and near-duplicates all remain candidates, so a random split would

TABLE I

Primary-variant data (ambiguous candidates excluded). Prevalence is the share of positive candidates; the effective sample size is the number of positive groups. IBM: τ quantile 0.05, maximum ring length 6, 5-day windows. Synthetic: τ 0.25, six seeds pooled per condition; "hn" = hard negatives, "chords" = dense_internal_chords.

Dataset	Candidates	Positives	Prevalence	Pos. groups
IBM HI-Small	7555	131	0.0173	14
IBM LI-Small	9720	38	0.0039	10
Synth. hn off, chords default	5071	103	0.0203	46
Synth. hn off, chords off	3860	103	0.0267	46
Synth. hn on, chords default	2477	102	0.0412	55
Synth. hn on, chords off	1933	93	0.0481	52

leak. Groups are formed once across all windows by union-find over candidates that share any edge, merged with the attempt (or planted structure) identifier. We use stratified group 5-fold cross-validation with 5 repeats and report the PR-AUC (average precision) of the pooled out-of-fold scores, averaged over repeats, next to the prevalence, since a random scorer attains the prevalence. Intervals are 95% percentile intervals of a grouped bootstrap that resamples groups (1000 resamples); comparisons between feature sets are paired differences on the same resamples. All scaling is fitted inside training folds.

A time-ordered split (train on rings first seen early, test on later ones) was tried as specified and was unusable: HI-Small left 0 test candidates and LI-Small 1, because rings recur and almost every window mixes early and late groups. It was dropped and no post-hoc variant was tried. The out-of-sample check is transfer between HI and LI, labelled as transfer: both come from one generator and cover the same dates, with 3 of 705,907 LI accounts shared with HI.

Synthetic results also carry the mean and standard deviation of the per-seed PR-AUC over the six seeds. Bootstrap draws come from a stored seed, so the same resamples are reused across feature sets and models, and the synthetic cells pool the six seeds with groups prefixed by seed so that seeds never merge. Hyperparameters, bootstrap size and random seeds are stored with every result file.

What is pre-specified. The primary comparison is B+D+T minus B+D; the secondary comparison is B+D minus B. Everything else, notably the T_{local} and T_{win} ablations (added after the first HI run), per-scenario rows, cap sensitivity, severity, clustering and the pump-and-dump track, is exploratory and labelled as such. Across the project 495 paired differences were computed: 55 primary, 47 secondary and 393 exploratory or sensitivity. No multiplicity correction is applied, and the rows are not independent; the headline test for each dataset is the primary comparison on the primary variant.

V. Results

All PR-AUC cells are pooled out-of-fold values with 95% grouped-bootstrap intervals; LR is logistic regression, RF random forest and SVM the RBF support vector machine. We first ask what carries the signal (V-A), then whether topology adds to it (V-B to V-D), and finish with ablations, candidate recall, the weight audit and the threshold study (V-E to V-I).

TABLE II

PR-AUC [95% grouped-bootstrap CI] on IBM AMLworld, primary variant. HI-Small: prevalence 0.0173, 14 positive groups. LI-Small: prevalence 0.0039, 10 positive groups. LR, RF, SVM as in the text.

	Features	LR	RF	SVM
HI	B	0.077 [0.033, 0.167]	0.080 [0.033, 0.157]	0.094 [0.040, 0.187]
HI	B+D	0.477 [0.240, 0.657]	0.652 [0.328, 0.830]	0.326 [0.126, 0.536]
HI	B+D+T	0.581 [0.300, 0.733]	0.676 [0.351, 0.868]	0.329 [0.135, 0.509]
LI	B	0.020 [0.007, 0.045]	0.010 [0.002, 0.031]	0.042 [0.008, 0.167]
LI	B+D	0.093 [0.030, 0.214]	0.142 [0.049, 0.306]	0.148 [0.030, 0.321]
LI	B+D+T	0.126 [0.032, 0.268]	0.130 [0.042, 0.259]	0.103 [0.036, 0.227]

A. Direction and behaviour features carry the signal

Table II and Fig. 3 give PR-AUC on the two IBM datasets. On HI-Small, B alone scores 0.077 to 0.094, about 4 to 5 times the prevalence of 0.0173 but far below B+D, which reaches 0.477 (LR), 0.652 (RF) and 0.326 (SVM). The paired gain of B+D over B (secondary, pre-specified) excludes 0 for all three models on HI-Small (LR +0.399 [+0.139, +0.571], RF +0.572 [+0.276, +0.720], SVM +0.233 [+0.057, +0.379]) and for two of three on LI-Small (LR +0.072 [+0.020, +0.173], RF +0.132 [+0.045, +0.283], SVM +0.105 [−0.060, +0.276]); see Table III. The synthetic generator agrees: the gain excludes 0 in every condition for every model (for example, hard negatives off with default chords, LR +0.440 [+0.324, +0.535]). On HI-Small, D alone matches or exceeds B+D for LR (0.581 against 0.477) and RF (0.682 against 0.652), but not for the SVM (0.131 against 0.326); these are unpaired comparisons with overlapping intervals (Fig. 3), so the supported claim is that D carries most of the signal, not that B adds nothing.

What stands out is how little structure alone buys. B sees degree, centrality and neighbourhood density, and it separates laundering rings from background loops poorly; what a ring's edges do (balance, regularity, birth) separates them far better.

B. No detectable topological lift on IBM data

With D established as the signal, the pre-specified question is whether T adds to it (Table III). On HI-Small (14 positive groups) the gain of B+D+T over B+D is +0.104 [−0.002, +0.171] for LR, +0.025 [−0.050, +0.102] for RF and +0.003 [−0.111, +0.077] for SVM. On LI-Small (10 positive groups) it is +0.033 [−0.007, +0.097] (LR), −0.012 [−0.119, +0.077] (RF) and −0.044 [−0.181, +0.091] (SVM). All six intervals include 0, and the point estimates change sign between the datasets for RF and SVM. With 14 and 10 positive groups, a small positive and a small negative effect are both compatible with the

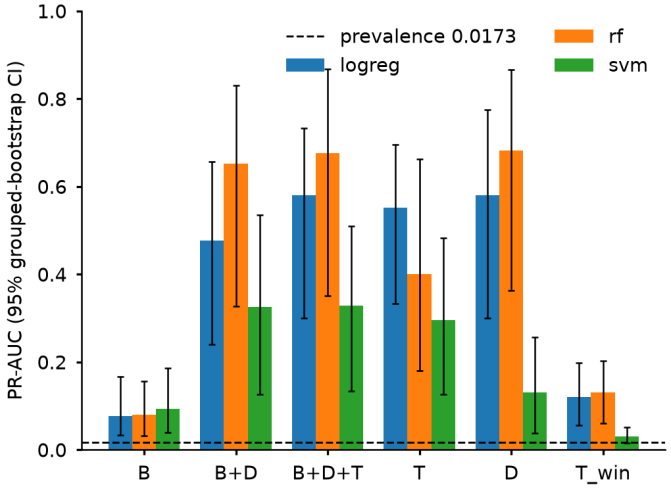


Fig. 3. IBM HI-Small, primary variant (ambiguous candidates excluded): PR-AUC of the pre-specified feature sets B, B+D, B+D+T and T, and of the controls D and T_{win} (exploratory), for logistic regression (logreg), random forest (rf) and RBF-SVM (svm). Prevalence 0.0173 (dashed line), 131 positives in 14 positive groups. Error bars: 95% percentile CI of a grouped bootstrap over groups. Pooled out-of-fold, mean of 5 repeats. Three exploratory ablations (B+D+T_local, B+D+T_win, T_local) are omitted; see text.

data, so this is a result about low power, not about absence of an effect.

It is not a ceiling effect. B+D leaves real headroom on HI-Small (0.326 to 0.652), yet topology moves PR-AUC by at most 0.104 in the primary variant.

The conclusion also survives a change in labelling. With ambiguous candidates counted positive, HI-Small has prevalence 0.0805 and 142 positive groups, and B+D+T minus B+D is +0.027 [−0.015, +0.086] (LR), +0.003 [−0.056, +0.032] (RF) and +0.176 [−0.033, +0.297] (SVM); LI-Small has prevalence 0.0513 and 189 positive groups, and the differences are −0.001 [−0.014, +0.008], +0.003 [−0.003, +0.008] and +0.007 [−0.009, +0.028]. All six intervals include 0.

C. Synthetic data: a small lift that does not carry over

The synthetic generator was built to give topology its best chance, and it does produce a gain, though a small one. B+D is already high there (0.960 to 0.983 for LR, 0.956 to 0.997 for RF and 0.874 to 0.937 for SVM over the four conditions), and D alone nearly separates the classes. B+D+T minus B+D has an interval above 0 in 7 of the 12 condition-model cells (Table III): all three models with hard negatives off, and only the SVM, with chords off, with hard negatives on (+0.052 [+0.001, +0.109]). For hard negatives off and default chords, B+D scores 0.970 (LR), 0.959 (RF) and 0.892 (SVM) and B+D+T 0.988, 0.996 and 0.969. Here the ceiling does matter: with B+D above 0.95 for LR and RF, the significant cells measure small gains in a thin margin. The gain does not appear on IBM, where no cell of the primary comparison (HI-Small, LI-Small or transfer) has an interval above 0. Two exploratory ablation cells on HI-Small do (B+D+T_{local} with LR, +0.141 [+0.031, +0.224]; B+D+T_{win} with SVM, +0.088 [+0.006, +0.176]), but the same ablations include 0 on LI-Small.

Where the gain falls is the more informative result (exploratory, per scenario; Fig. 4). We expected topology to be nearly redundant for isolated rings and to matter for rings whose loops are filled, split or crowded by their neighbourhood. The counts of model-condition cells with an interval above 0 say otherwise: isolated rings 8 of 12 (SVM in all four), dense rings 2 of 12 (only with internal chords off and hard negatives off: LR +0.130 [+0.038, +0.330], RF +0.123 [+0.029, +0.329]), and chorded rings 6 of 12 (SVM 4 of 4, RF 2 of 4, LR 0 of 4). The isolated result is the surprising one. For an isolated ring the H1 class is born at the ring's birth value and never dies (Section III), so T should repeat what D already holds, and a gain there suggests that T supplies something else. Since T also encodes neighbourhood size and degree structure, one reading, untested, is that it contributes context rather than loop topology. The dense result has a simpler partial explanation: with default chords only 7 of 24 planted dense rings keep an open loop, so for most dense positives persistence reports a filled loop while the label stays positive, and the only dense cells with a gain are those with chords off. The per-scenario groups are few (39 isolated, 10 dense, 8 chorded at hard negatives off with default chords), so these counts are suggestive at most. When the planted shortcut-filled pseudo-rings are ranked, adding T lowers their mean out-of-fold percentile under logistic regression (B+D 0.982, B+D+T 0.909, hard negatives off; descriptive means, no interval), the one place where the intended direction appears.

D. Transfer and the ring-birth signal

We next ask whether anything learned on one IBM dataset carries to the other. Training on one and testing on the other (labelled transfer; one model per direction) gives B+D PR-AUC of 0.133, 0.149 and 0.124 for LR, RF and SVM from HI to LI at prevalence 0.0039. B+D+T minus B+D is again not above 0 in any cell (Table III); it is below 0 for RF from HI to LI (−0.066 [−0.177, −0.012]) and for LR from LI to HI (−0.276 [−0.550, −0.036]), so there topology hurt rather than helped. Ring birth by itself, ranked without any model (−D_birth), scores 0.314 [0.097, 0.487] on HI-Small at prevalence 0.0173 but 0.006 [0.002, 0.025] on LI-Small, which cannot be told apart from that dataset's prevalence of 0.0039. The signal seen on HI-Small therefore does not replicate on LI-Small. These are separate within-dataset scores of a model-free ranking, not a transfer result. The weight audit (V-G) gives a partial reading: the weights separate IBM cycles mostly through balance, and ring birth is built from them. Why that helps on HI-Small and not on LI-Small we cannot say.

TABLE III

Paired PR-AUC differences [95% grouped-bootstrap CI over groups, same resamples for both feature sets]. Primary (pre-specified): B+D+T minus B+D. Secondary (pre-specified): B+D minus B. HI-Small: 14 positive groups; LI-Small: 10; synthetic conditions: 46, 46, 55 and 52 positive groups (Table I); transfer: one model per direction, CI over test groups (HI→LI: 10; LI→HI: 14).

Dataset	Comparison	LR	RF	SVM
HI-Small	primary	+0.104 [−0.002, +0.171]	+0.025 [−0.050, +0.102]	+0.003 [−0.111, +0.077]
LI-Small	primary	+0.033 [−0.007, +0.097]	−0.012 [−0.119, +0.077]	−0.044 [−0.181, +0.091]
Transfer HI→LI	primary	−0.017 [−0.147, +0.092]	−0.066 [−0.177, −0.012]	−0.050 [−0.142, +0.042]
Transfer LI→HI	primary	−0.276 [−0.550, −0.036]	−0.019 [−0.079, +0.046]	−0.056 [−0.123, +0.025]
Synth. hn off, chords default	primary	+0.018 [+0.002, +0.057]	+0.038 [+0.010, +0.095]	+0.077 [+0.019, +0.175]
Synth. hn off, chords off	primary	+0.030 [+0.010, +0.074]	+0.043 [+0.011, +0.110]	+0.096 [+0.029, +0.210]
Synth. hn on, chords default	primary	+0.005 [−0.009, +0.021]	−0.003 [−0.012, +0.004]	+0.038 [−0.004, +0.080]
Synth. hn on, chords off	primary	+0.014 [−0.006, +0.039]	+0.003 [−0.004, +0.012]	+0.052 [+0.001, +0.109]
HI-Small	secondary	+0.399 [+0.139, +0.571]	+0.572 [+0.276, +0.720]	+0.233 [+0.057, +0.379]
LI-Small	secondary	+0.072 [+0.020, +0.173]	+0.132 [+0.045, +0.283]	+0.105 [−0.060, +0.276]
Synth. hn off, chords default	secondary	+0.440 [+0.324, +0.535]	+0.383 [+0.265, +0.477]	+0.411 [+0.213, +0.520]
Synth. hn off, chords off	secondary	+0.414 [+0.290, +0.509]	+0.389 [+0.248, +0.519]	+0.368 [+0.184, +0.487]
Synth. hn on, chords default	secondary	+0.542 [+0.418, +0.621]	+0.389 [+0.258, +0.494]	+0.598 [+0.458, +0.676]
Synth. hn on, chords off	secondary	+0.510 [+0.376, +0.600]	+0.395 [+0.244, +0.513]	+0.614 [+0.468, +0.698]

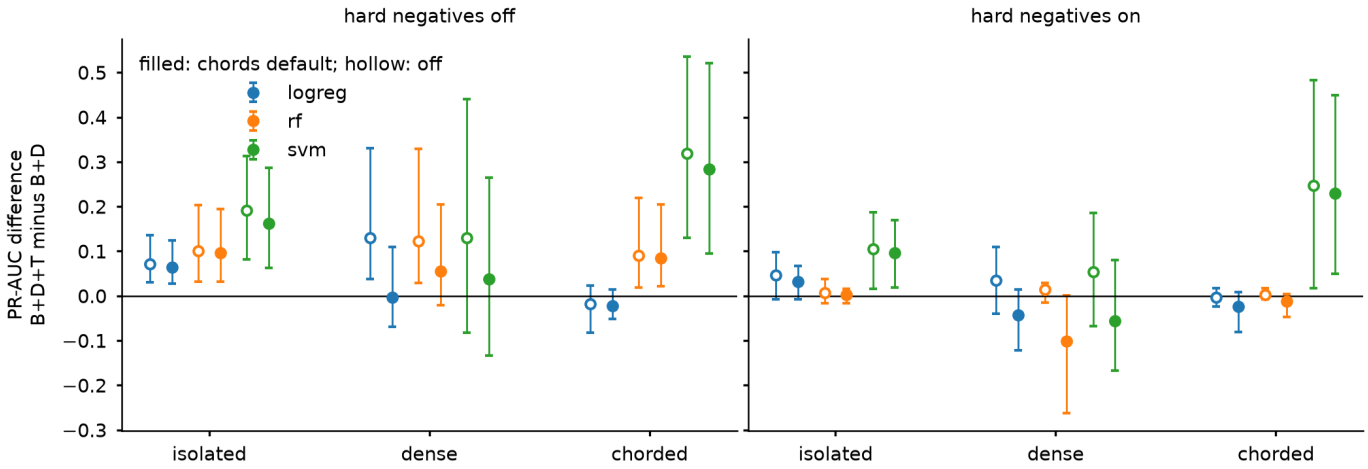


Fig. 4. Exploratory. Synthetic data, six seeds pooled: paired PR-AUC difference B+D+T minus B+D per scenario for logistic regression, random forest and RBF-SVM, hard negatives off (left) and on (right); filled markers: dense_internal_chords default, hollow: off. Error bars: 95% percentile CI of a grouped bootstrap over groups (same resamples for both feature sets). Each scenario's positives are compared with all negatives, so prevalence differs per scenario: isolated 0.0108 to 0.0265, dense 0.0048 to 0.0113, chorded 0.0050 to 0.0118; positive groups per scenario at hard negatives off with default chords: isolated 39, dense 10, chorded 8.

E. Exploratory ablations

Splitting T into its local and window parts is informative. With HI-Small, B+D+T_{local} minus B+D is +0.141 [+0.031, +0.224] for LR, +0.041 [−0.037, +0.106] for RF and +0.037 [−0.081, +0.126] for SVM, while B+D+T_{win} minus B+D is +0.034 [−0.005, +0.094], +0.007 [−0.025, +0.059] and +0.088 [+0.006, +0.176]. These ablations were added after the first run and are exploratory, one model each. The control is more telling: T_{win} alone, constant within a window, still scores 0.121 [0.056, 0.198] (LR) and 0.130 [0.061, 0.202] (RF) on HI-Small at prevalence 0.0173, against 0.006 and 0.004 on LI-Small at prevalence 0.0039. A feature that cannot distinguish rings within a window can only be identifying windows, and laundering attempts are concentrated early in the IBM data, so window identity can carry label information. We did not test this reading. Among the infinite-bar caps fixed in advance (window maximum, fixed 1.0, 1.1 times the window maximum) the largest absolute paired difference to the default is 0.024 (SVM, B+D+T, HI-Small; fixed 1.0 against the window maximum, −0.024 [−0.037, −0.004]). A cap based on the local maximum, examined after seeing results, changes the HI-Small SVM B+D+T value by +0.190 [+0.097, +0.240]; it is post hoc, from one model, and not adopted (Appendix B).

F. Candidate recall

Candidate generation caps what any classifier can find (Table IV). In the first HI-Small window (days 0–5), 15 CYCLE attempts are complete, 11 involve at most 6 accounts, and 8 of those are recovered at τ quantile 0.05 (6 also carry a positive CYCLE label under the 80% rule). On the whole HI-Small data, 19 of 34 reachable attempts are

TABLE IV
Candidate recall of complete CYCLE attempts on IBM data, τ quantile 0.05, maximum ring length 6. "Reachable" means at most 6 accounts. "Recovered" means a candidate with exactly the attempt's account set; "labelled" adds that the candidate carries a positive CYCLE label. Counts, not a sample: no interval. LI days 0–5 contains one complete attempt.

Data	Scope	Complete	Reachable	Recovered	Labelled
HI	days 0–5	15	11	8	6
HI	whole data	54	34	19	15
LI	whole data	12	6	3	3

recovered. On the synthetic generator with hard negatives off, recall of planted rings is 1.00 at every τ tested; with hard negatives on it is lower at $\tau = 0.05$ (0.52 isolated, 0.21 dense, 0.42 chorded), 1.00 at 0.15 to 0.25, and limited by the enumeration cap above.

G. Audit of the weights on IBM patterns

The weights were designed for wash trading, so we checked how they behave on IBM patterns. For HI-Small windows 0 and 2 pooled, the edge-level AUC of CYCLE pairs against background pairs is 0.794 for the combined weight, which is built from rank-normalised scores. For the raw scores it is 0.795 for balance and 0.601 for timing; for the rank-normalised scores it is 0.804 and 0.589. The RANDOM pattern type has a similar weight AUC (0.785), and the other attempt types range from 0.528 to 0.653. About 0.899 of CYCLE pairs have fewer than three payments, so their timing score is the imputed median, against 0.467 of background pairs. The wash-trading weights therefore describe IBM cycles only partly, mostly through balance, which is why IBM is used to test the machinery.

H. Adaptive thresholds

On the drift stream of the optimization report (40 windows, six stream seeds, $\lambda = 1$, lag $d = 2$), the share of the hindsight-oracle reward is 0.298 ± 0.059 for a fixed threshold, 0.861 ± 0.020 for re-tuning every 5 windows, and 0.981 ± 0.006 for online gradient ascent (Table V). The bandits lose: discounted UCB is -1.245 ± 1.066 below the re-tuned comparator (0 of 6 seeds better) and sliding-window Thompson sampling -0.771 ± 0.744 below (2 of 6). A plausible reason, which we did not test, is that 40 windows with delayed labels leave a bandit little time to explore its arm grid, while re-tuning uses recent labelled history directly. The only positive cell, discounted UCB at lag 5 (+0.258, all 6 seeds), is not a skill result: on stream seed 0 that policy never played the last arm within its 35 windows. A single play of the arm that flags almost everything costs about as much as the whole oracle total, and rewards are clipped at ± 10 before the policies see them, which hides its size. Hyperparameters were set in advance; a later run with a larger discount did better and was not adopted. This is one simulated stream and does not show that bandits are poor in general.

TABLE V
Threshold policies on the synthetic drift stream of the optimization report: share of hindsight-oracle cumulative reward at $\lambda = 1$, lag $d = 2$ (mean $\pm$ std over 6 stream seeds; not a CI), and paired difference of the bandits against re-tuning every 5 windows (retune_5), with the number of seeds on which the bandit is better. Synthetic only.

Policy	Share of oracle	Minus retune_5
Fixed	0.298 ± 0.059	
Re-tune every 5 windows	0.861 ± 0.020	
Online gradient ascent	0.981 ± 0.006	
Discounted UCB	-0.383 ± 1.067	-1.245 ± 1.066 (0/6)
Sliding-window Thompson	0.091 ± 0.753	-0.771 ± 0.744 (2/6)

I. Other exploratory studies

Three further studies are exploratory and summarized in Appendix B. A next-window severity regression on synthetic rings (IBM has no next-window wash volume) recovers generator structure: ring length alone has a root-mean-square error of 0.132 [0.125, 0.139] for random forest against 0.141 [0.134, 0.149] for B+D, and T adds nothing. Ward clustering of positive IBM rings agrees only weakly with pattern types (HI-Small, $k = 4$: adjusted Rand index 0.097 against 0.398 against group identity), so the clusters track attempts more than types; the bootstrap intervals for this study are biased and are not used. A synthetic pump-and-dump track with no topology shows that the harness runs on a second pattern type, and the generator is nearly separable by exit timing.

VI. Discussion and Limitations

What the evidence supports. The consistent finding is that what a ring's edges do matters more than where the ring sits. Direction and behaviour features, which include the ring birth that the filtration produces, carry the detection signal on every dataset we tried, and topological features add nothing we can detect on top of them within either IBM dataset. The synthetic generator, built so that topology should matter, gives a small gain that is not concentrated where the design predicted. We therefore make no claim that topology improves detection on real wash-trading data, and the results do not say that it fails to: this experiment cannot tell.

Reading the null result. The width of the IBM intervals is set by the number of positive groups, not the number of positive rings: 131 positive candidates give only 14 groups. The largest upper bound among the within-dataset primary intervals is +0.171 (logistic regression, HI-Small), so a gain of that size cannot be ruled out, and neither can a small loss. In transfer, the two cells whose intervals exclude 0 are negative. Three explanations for why topology adds little fit the design, and all are untested. Ring birth, which the filtration produces, already sits in D, and it is what an isolated ring's H1 bar reports. The baseline sees the full window graph, so neighbourhood context is not exclusive to T. And for isolated rings persistence repeats

enumeration. The window-level topology set is a separate matter: it is where a time proxy would act.

Cost. Whether persistence features justify their computational cost is not settled by this study; we did not measure a cost-benefit trade-off.

Limitations.

1) Few positive groups. The effective sample size is the number of positive groups: 14 on HI-Small (131 positive candidates) and 10 on LI-Small (38). In 144 of 150 fold-repeats over three τ values, the two datasets, five folds and five repeats, a fold holds fewer than 5 positive groups. At this size a single-dataset signal is fragile, as the ring-birth detector shows (it works on HI-Small, not on LI-Small). LI-Small has only 3 CYCLE positives, so a cycle-only analysis is unusable there.
2) Time proxy. Window-constant features alone score above chance on HI-Small (V-E); the explanation is untested.
3) Synthetic data. Both IBM datasets and our own generator are synthetic; with hard negatives on, 0.963 (chords default) and 0.972 (chords off) of all candidates are ambiguous.
4) Weights and imputation. The scores encode wash-trading behaviour, and 0.685 to 0.966 of each laundering group's pairs have an imputed timing score. IBM amounts mix currencies. IBM data are not market transactions.
5) Candidate recall. Only 11 of 15 complete HI cycle attempts in the first window can be reached with at most 6 accounts, and the bounded-enumeration algorithm has a published caveat [15].
6) Sampled betweenness. On IBM windows betweenness uses 100 sampled sources, so most nodes score 0 (0.1594 of the 404,273 nodes of the first HI window are non-zero).
7) Untuned SVM and no GNN. The SVM has fixed hyperparameters and is weaker than the random forest on HI-Small (B+D 0.326 against 0.652). No graph neural network, community-detection or published topological method was run, so we make no claim of outperforming any of them.
8) Multiplicity. 495 paired rows were computed with no correction; the exploratory ablations in particular should be read as hypotheses.
9) Threshold study. One simulated drift stream, a fixed arm grid and reward clipping.

What was not done. The planned NFT experiment on real wash-trade labels was not run: running queries on Dune is blocked on its free tier. BigQuery access was confirmed and the schemas inspected; the planned leakage-controlled subset would use only the proxy-label filters that are not cycle rules, because two of the five filters of the public Dune wash-trade model (as inspected by us) are cycle-type rules and would leak the structure our method detects. We also did not run an undirected clique-complex ablation, so the benefit of the directed construction is argued from construction (and checked on toy graphs), not measured; a persistence-image ablation; classifiers on single-score filtrations; or sweeps over window length, stride, label threshold and, for classification, τ and maximum ring length (only candidate recall was swept). A real-data pump-and-dump experiment is likewise absent. Finally, the time-ordered evaluation planned in the design was unusable (Section IV), so the out-of-sample evidence is transfer between two datasets from the same generator.

Deviations from the design. De-duplication keeps the shortest ring first and then the lowest birth, and removes nothing for lengths up to 9; the optimization report keeps a maximum length of 5 and a fixed cap of 1.0 for its legacy topological columns while the classification uses 6 and the window-maximum cap; timing is imputed for pairs with fewer than three payments, not only for a single payment; a cycle-only subset and the ambiguous-counted-positive variant were added; and the planned false-positive rate at 80% recall was replaced by the rank percentile of each negative kind.

Ethics. A ring score is a suspicion score, not a finding of guilt. Publishing the scores may help adversaries who inject decoy trades to break balance or regularity.

VII. Conclusion and Future Work

We built a pipeline that localizes candidate rings exactly and describes each by direction and behaviour features, graph baseline features and persistent homology of a directed flag complex, and we compared B+D+T with B+D under a protocol fixed in advance. Direction and behaviour features carry the signal. Topology adds no detectable lift within either IBM dataset, with intervals that cannot exclude a small effect of either sign, and two of six transfer cells are negative. On a nearly separable synthetic generator it adds a small lift that does not carry over. Future work includes the NFT experiment on the cycle-free subset of proxy labels, an undirected clique-complex ablation, persistence-image and learned-vectorization ablations, sweeps over the parameters listed above, more positive groups through larger IBM variants, and a direct test of whether the window-level features act as a time proxy.

Appendix A
Optimization-report results (pinned generator)

These numbers come from the optimization report on its pinned synthetic generator (dense, chorded, shortcut-filled and hot-wallet counts set to 0; ambiguous candidates counted as negatives), not from IBM data. Ring-birth detector test average precision rises from 0.5675 with equal weights to 0.9997 with weights learned on the simplex, (0.0, 0.2067, 0.7933) for (reciprocity, balance, regularity); learning matters because of Proposition 1. A logistic classifier on D reaches test average precision 0.9984 and with

TABLE VI
Exploratory side studies. Severity and pump-and-dump are synthetic only. Clustering uses IBM HI-Small positive rings (131 rings in 14 groups; $k = 4$, D plus T_{local} landscape). Percentile CIs of the clustering bootstrap are biased and not used. No multiplicity correction.

Study	Result
Severity, RMSE (random forest)	B+D 0.141 [0.134, 0.149]; D_{length} alone 0.132 [0.125, 0.139]; fair persistence fit 0.221 [0.203, 0.237]; B+D+T minus B+D −0.001 [−0.004, +0.002]; random-forest importance of D_length 0.796
Clustering, HI	ARI against type 0.097 (random 0.000); against group id 0.398; purity 0.489 (random 0.365, majority 0.336)
Clustering, LI (T_{local} landscape, $k = 3, 9$ groups)	ARI 0.260; purity 0.730 (random 0.536, majority 0.459)
Pump-and-dump, default exits (47 positive groups, prevalence 0.390)	M features 0.890 (LR), 0.976 (RF), 0.928 (SVM) against group-size baseline 0.388, 0.375, 0.414; −M_exit_spread_h alone 0.906 [0.860, 0.945]; with wider exits 0.643 [0.513, 0.752]
Cap, HI SVM B+D+T	window maximum 0.329; fixed 1.0 0.305; 1.1× maximum 0.334; local maximum (post hoc) 0.518

the legacy topological columns 0.9969, so topology does not help the report's detector either. Gradient descent with a step matched to the local curvature reaches an optimality gap of 10^{-6} in 322 iterations. On the 40-window drift stream, cumulative reward after warm-up as a share of the oracle is 0.2367 for a fixed threshold, 0.86 for re-tuning every 5 windows and 0.97 for online gradient ascent. Seed 0 of the bandit benchmark reproduces the report's stream exactly (fixed 71.0, re-tune 258.0, ascent 291.0, oracle 300.0).

Appendix B
Exploratory results

Table VI lists the exploratory side studies, with values taken from the stored result files; severity and pump-and-dump are synthetic only.

Appendix C
Reproduction

Every number in this paper was read from stored result files by a script that records its source; the stored seeds are cross-validation seed 0, bootstrap seed 12345, model seed 0 and synthetic seeds 0 to 5. The pre-existing optimization-report outputs are never overwritten. The test suite collects 220 tests; model performance is never a test.

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